

Mental Skills for Ultrarunning

Don't let the relatively slow pace fool you—ultrarunning is a physically torturous sport. With every step, our feet impact the ground with force a few times greater than our own body weight. This collision with the earth reverberates from our feet through our legs, into our torso, and finally to the neck, shoulders, and head. Over the course of an ultramarathon, we take tens of thousands of steps, resulting in muscle damage so great it can be comparable to the physical trauma of a serious accident or illness. We are reminded of the physical brutality of the sport after races as we hobble around, unable to stand, sit, or walk without grimaces, grunts, and assistance. But the physical damage does not stop at the muscles. During an ultra, our gastrointestinal system is also damaged. The sheer volume of foods and fluids consumed, as well as all the jarring and jostling of running can cause physical damage to the gut that is characteristic of an acute infectious episode. Still, damaged muscles and a dysfunctional gastrointestinal system are not the end to the damage train. Ultrarunners also suffer injuries to their musculoskeletal system during the course of training and racing. Regardless of whether it's spraining an ankle, catching your toe on an unsuspecting rock and slamming onto the ground superman style, or suffering an overuse injury, these maladies are physical reminders of our interactions with the environment.

The volume and severity of physical insults are so great that we tend to give them our undivided attention. We want big strong quadriceps to withstand the torturous nature of downhill running. We want voluminous lungs and a powerful heart in order to deliver oxygen and blood to working muscles for hours on end. We measure and record things like cardiac output, VO₂ max, vertical gain, and mileage as physical reminders of the work we have done. The size, shape, and volume of these reminders serve as measuring sticks for improvement, the bigger the better. But the intense focus on the physical exposes our biggest blind spot as athletes: mental training.

Athletes generally give lip service to the idea that ultrarunning is a mental sport. Clichés like “the most important distance is the 6 inches in between your ears” and “running is 90 percent mental and the other 10 percent is mental” are found in magazines and podcasts and thrown around like cheap confetti. We revere the ideas of “grit” and “toughness” even though there is no way to encapsulate them on your Strava dashboard. We even get to see the manifestations of an athlete's mentality as they forge on after hours of not being able to eat, with macerated feet and with bloodied knees, leaving us to wonder, “How did she finish?”

Despite the clichés and visual reminders of the mental side of ultrarunning, most notions of training the mental aspects of the sport unfortunately stop there. We recognize that the space in between your ears is important, that athletes with better mental skills will outperform their physiologically equivalent peers, and that mental skills training can improve race outcomes, but we very rarely take the steps to make those improvements. Add that to the general dearth of mental skills training programs and professionals in the space (heck the first edition of this book missed this subject altogether, and many athletes do not have the first clue where to start).

Unlike traditional physiological improvements, mental skills development is difficult to measure directly, and progress is hard to track. With physiological improvements we can directly measure parameters like oxygen consumption, determine how changes in oxygen consumption relate to performance, and then design and deploy training interventions to test their ability to improve performance. The same is true with muscle strength, force production, and the formation of lactate and other metabolites: we can measure them, learn about their importance to performance, and then design interventions that hopefully impact those variables. As a result of all of this measuring, testing, and remeasuring, we know a tremendous amount about the physiology that drives effective interval design and training structure.

The causal relationship between developing mental skills and improving physical performance is less concrete, which unfortunately encourages an “if you can't measure it, it's not worth doing” way of thinking. Because it's difficult to directly tie the application of mental skills to specific physical improvements, coaches and athletes stick to what they can reliably measure. Indeed, much of the research on mental skill development is mechanistic in nature, rarely yielding clear performance outcomes. For example, we can trace the impact of mindfulness on

brain function by using functional MRI images to see what areas of the brain exhibit greater activation when meditating. Yet the link from that brain activation to performance is what I would describe as an “extension of the mechanism.”

On the whole, research findings in this area are equivocal between mental skills development having some effect and no effect on performance. Because of this conundrum, the first part of this chapter will be a primer on how your psychology interacts with your physiology to impact performance. My intention is to show you, using research, logic, and examples, that mental skills either enhance or limit your performance. After that, I will walk you through a few mental skills you can use during your day-to-day training and teach you to logically string them together, much the same way you would organize your physical training. In keeping with the rest of this book, there will be no golden answer! Just as I have intentionally omitted static physical training plans from this book, you will find no static program for mental skills development in this chapter. There are, however, common themes, strategies, and a general framework you can apply to your training to help you become a better ultrarunner.

Psychobiological Model of Fatigue

Earlier in this book I presented the concept of limiting factors in ultrarunning. These limiting factors included such things as musculoskeletal stress, thermal load, glycogen depletion, and the like. Undoubtedly all of these physiological phenomena impact performance. However, there’s a growing understanding that what slows us down is not necessarily the physiological manifestation of any of these limiting factors; rather, it’s how we psychologically perceive them. In this sense, our physiological capacity determines the upper limit of performance capacity, but our psychological skills are the deciding factor in how much of that capacity we ultimately utilize (Ikai and Steinhaus 1961). Figure 15.1 graphically illustrates this concept, as well as how improved mental skills can help you realize the most of your physiological capacity.

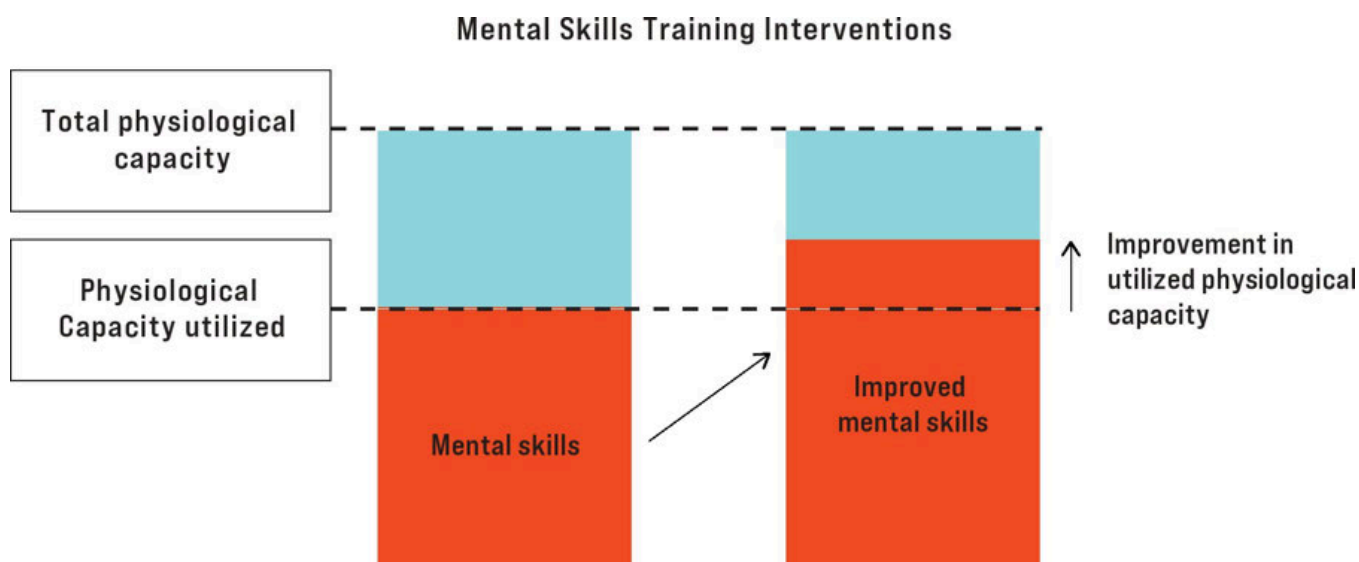


FIGURE 15.1 How improved mental skills can help you utilize more of your total physiological capacity.

This integrated model of physiology and psychology has been dubbed the psychobiological model of fatigue (PBM). In this sports psychology construction, an athlete’s motivation and perception of effort are the primary drivers of performance. It differs from other psychological models, most notably Tim Noakes’s central governor theory (Noakes 1997, 2012; Noakes et al. 2005). PBM recognizes that the brain and body act in concert to regulate performance, constantly modulating aspects of effort, pain, fatigue, and our perception of these

sensations, whereas the central governor theory, as it was originally conceived, centered on the brain's role in protecting against catastrophe as the ultimate regulator of performance (though nuanced iterations of Noakes's model subtly changed this viewpoint over the years).

The difference is not inconsequential. Noakes's central governor theory designates the brain as the ultimate regulator whose only goal is to guard against catastrophe. Flowing from this theory, coaches and athletes have used psychological and physiological training strategies that were anchored around maximizing an athlete's ability to tolerate pain and discomfort. The "no pain, no gain" ethos found in many training strategies throughout the early 2000s paid homage to this. High-intensity training was all the rage for endurance athletes. Runners, cyclists, and triathletes alike pushed hard—often too hard—simply for the sake of improving the maximal amount of effort the regulator (brain) would tolerate before initiating shutdown. Athletes were taught to be "tough" and "gritty" and to persevere by whatever means necessary to achieve those ends. While toughness and grittiness are desirable traits for any athlete, they are not the only components of a comprehensive set of mental skills.

As opposed to the central governor theory's focus on the brain as the final arbiter of performance regulation, PBM puts the athlete and her regulatory skills in the driver's seat. PBM is anchored in the understanding that we psychologically interpret the physiological manifestations of things like pain, effort, drowsiness, fatigue, and discoordination and thus have the ability to voluntarily change regulatory strategies like pacing to optimize performance. PBM leaves room for the athlete to deliberately enhance aspects of motivation, consciously shift attention inward and outward, and even consume ergogenic aids like caffeine to alter the way we perceive the sensations that are telling us to slow down.

Perceived Exertion End Point Interactions

PBM can also explain why athletes drop out of races when there is plenty of physiological reserve to continue, through what is known as the "perceived exertion end point interaction." During any difficult task, we routinely and consistently monitor our effort in an attempt to correctly mete out our effort and achieve the best result (Figures 15.2 and 15.3). Think of an interval workout consisting of three by twelve-minute TempoRun intervals, for example. Typically, you pace the first interval in such a way that you can complete the next two. If, during the middle of the first interval, you think you are going too hard, you adjust your effort accordingly. In order to make these adjustments, you are drawing in your mind a line from your current effort level to what you think you can tolerate at the end of the task (the perceived exertion end point). As the workout goes on, you interpret new sensations of pain and fatigue and redraw this line over and over again, leading you to modulate your effort all the way to the end.

This process works very well in workouts and short races. We have a good sense of our maximum tolerable exertion level and how we can get to that maximum tolerable exertion over the course of a set of intervals, a 5K, or even a marathon. However, in an ultramarathon, this strategy backfires (Figure 15.4). Our predictive capabilities can only extend so far, and the sheer duration of an ultramarathon confounds our ability to accurately predict how this internal struggle will ultimately play out. As a consequence, you see athletes drop out halfway through an ultramarathon saying "I don't think I can finish" only to arrive at their hotel room moments later and immediately regret the decision. On the flip side, you can leverage knowledge of PBM and perceived exertion end point interactions to enhance ultramarathon race performance by "staying in the moment," which, although cliché, is an effective tool for bypassing your body's natural tendency to evaluate and predict effort (Figure 15.5). This will be discussed in more detail in the following chapter.

The message I want you to come away with is that your ultramarathon performance will be determined by a combination of your physiological capacity and your mental skills. Both can be enhanced by training. Your maximum capacity for performance is determined by your physiology, but how much of that maximum you actually utilize is determined by your mental skills.

Author's note: This interpretation of PBM is an amalgamation of several authors' descriptions of the phenomena, most notably Samuele Marcora. For further reading, please see Marcora 2008, Marcora 2009, and Marcora and Staiano 2010. Full references can be found at the back of the book.

Associative and Dissociative Focus Strategies

There is a lot of space in the sport of ultrarunning. There is linear space for miles on a ribbon of singletrack or miles of trail or road. There's the vertical space the mountains occupy as they jut out from the landscape. Finally, there is also the space in time ultrarunners experience, stemming from the duration of the event and various training activities. The last of these creates a lot of space within an ultramarathoner's mind, which can be filled with any number of things. We can calculate the time to the next aid station, focus inwardly on how our body feels, distract ourselves with music, take in the scenery, and occupy our wandering minds with any number of productive or counterproductive thoughts.

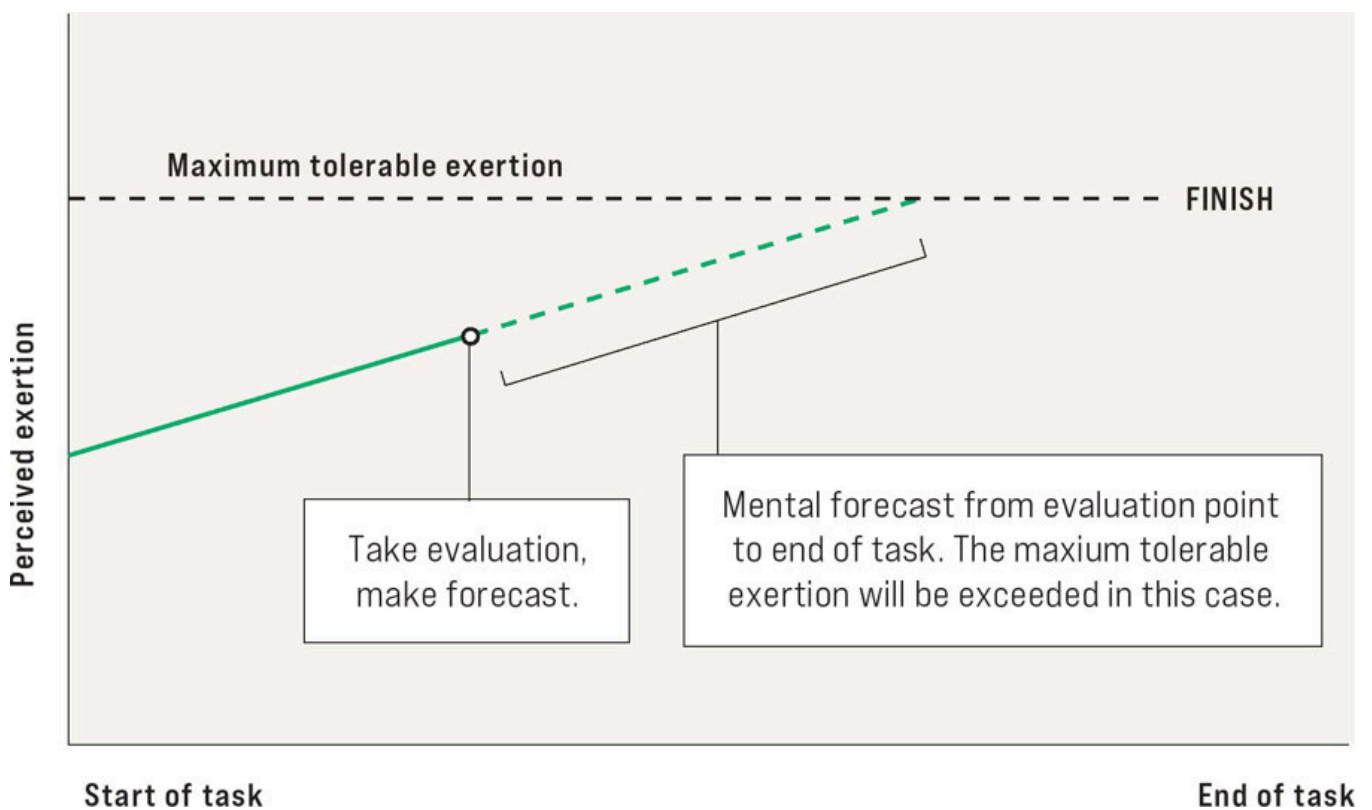


FIGURE 15.2 Schematic of the perceived end point interaction.

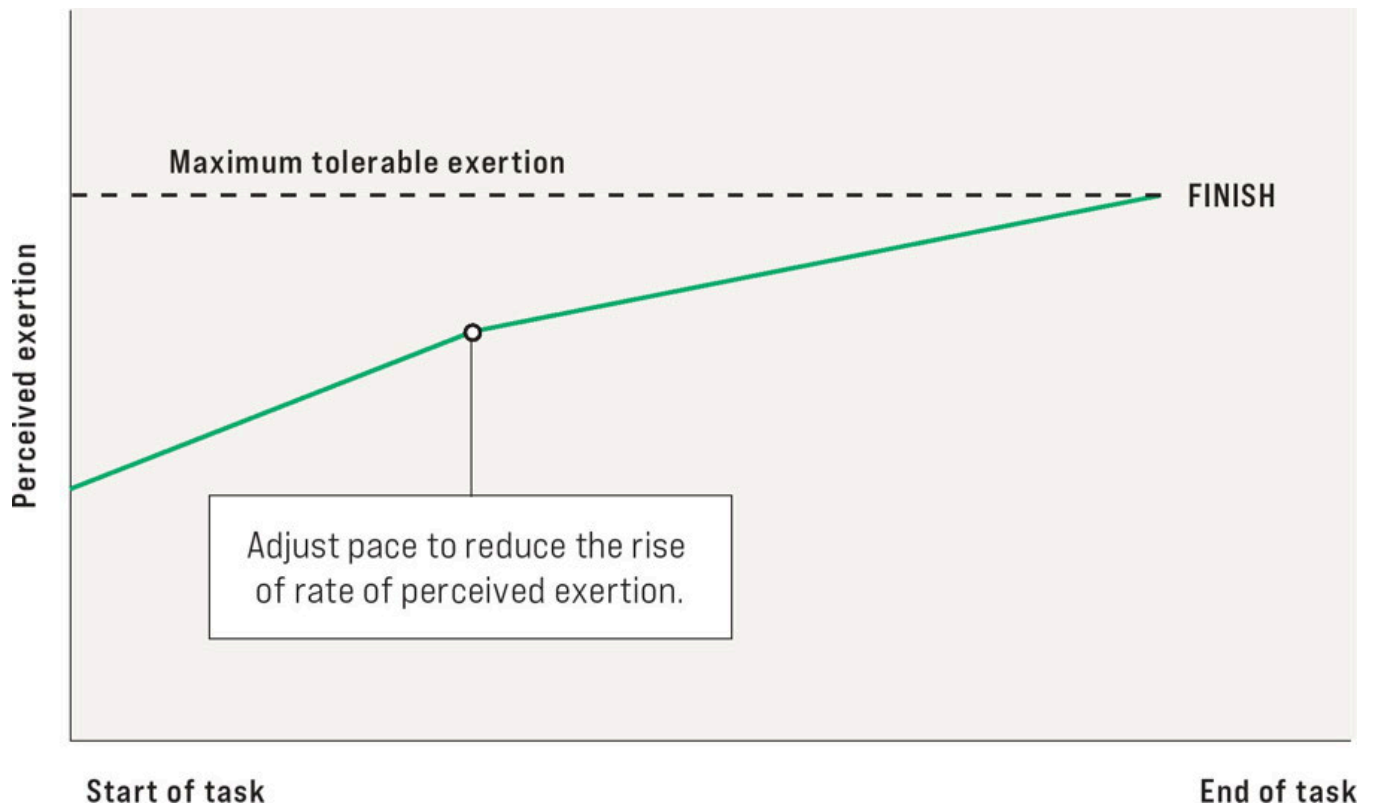


FIGURE 15.3 How an athlete adjusts pacing due to perceived end point interactions.

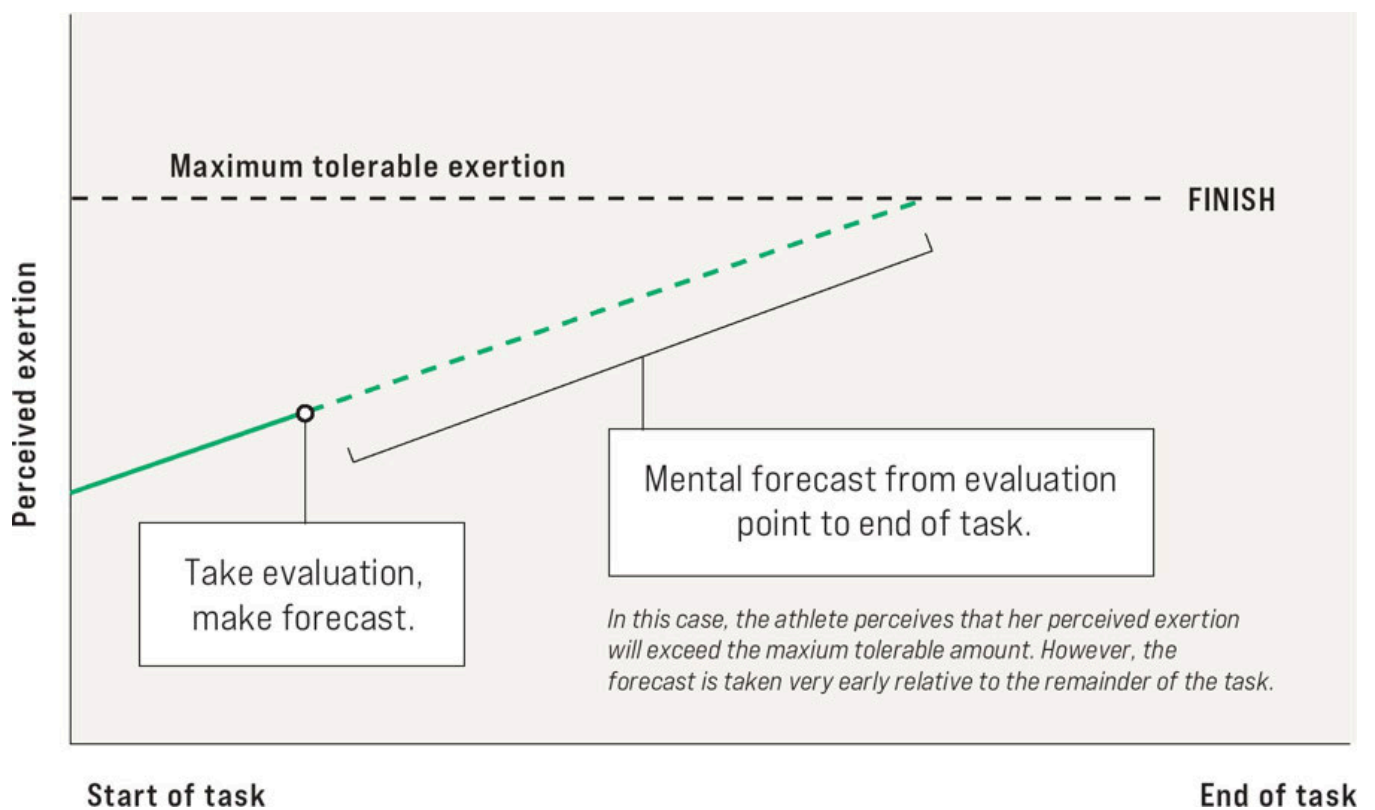


FIGURE 15.4 How an athlete can go awry with an early perceived end point interaction forecast.

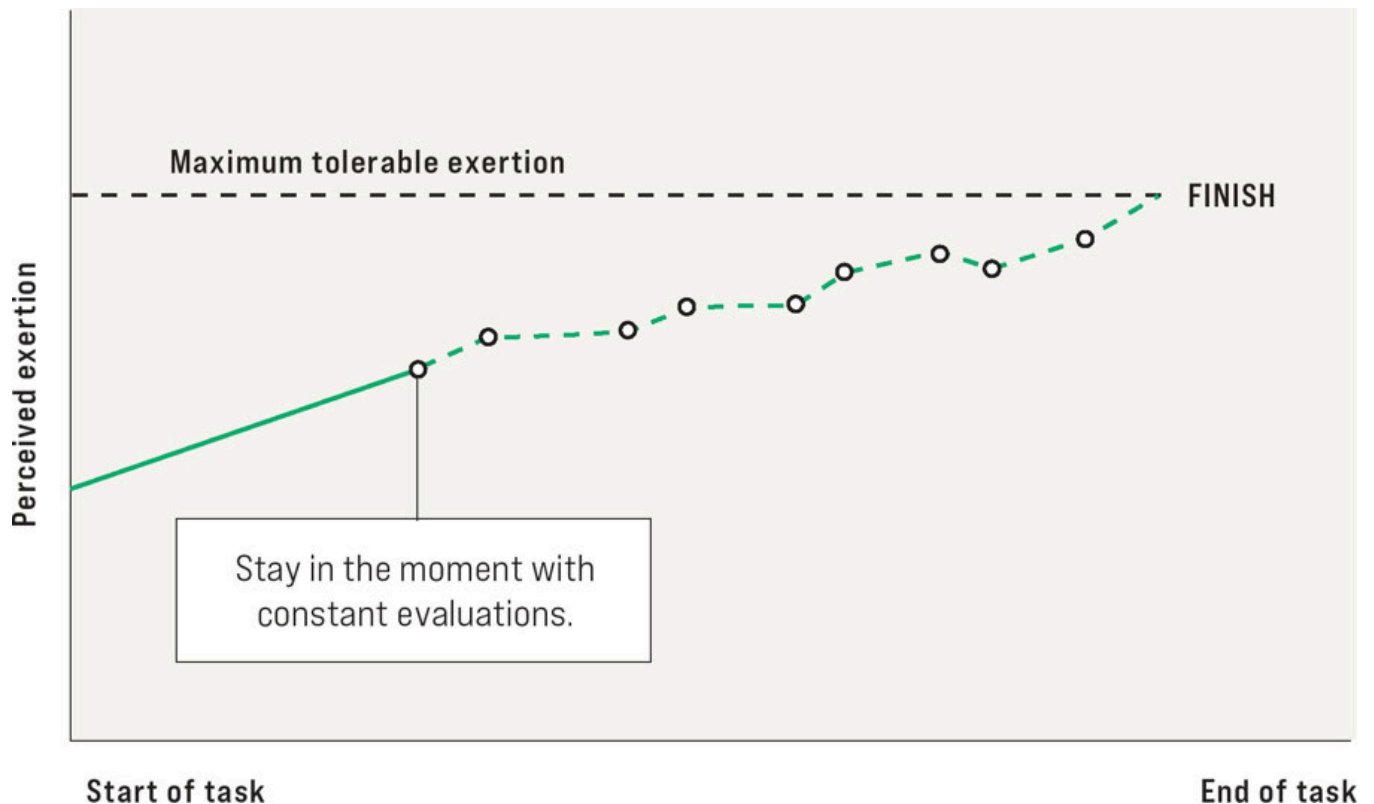


FIGURE 15.5 How staying in the moment avoids inaccurate forecasting.

In 1977, seminal research performed by Michael Pollock and William Morgan on focus and cognitive strategies revealed that, throughout the course of an event, elite marathon runners were more likely to focus on internal cues such as breathing rate and how their legs felt than their non-elite counterparts (Morgan and Pollock 1977). Nonelite athletes tended to turn their attention outward, choosing to distract themselves by thinking about uncompleted home projects, memories of the past, and other events unrelated to running as a way to distract them from the effort at hand. This bifurcation of focus has led the sports psychology field to frame focus strategies into two different categories, termed “associative” and “dissociative.” As the adjectives imply, athletes using associative strategies direct their focus inward toward sensations and feelings they are presently experiencing. These strategies harness information derived from physiological sensory feedback in order to regulate performance. Dissociative strategies take the opposite approach, wherein the athlete temporarily removes himself from the present situation and redirects thoughts to something unrelated to the task. This can be as simple as counting or as complex as planning an elaborate meal, all in an effort to occupy the mind with anything other than what is happening with your physiology.

While this formative research created a distinction between elite athletes who focused inwardly and non-elite athletes who focused elsewhere, interventions should not be exclusively aimed at improving associative focus to be more like the elites, as Morgan and Pollock’s original research implied. In the time since that research was performed, athletes, coaches, and sports psychologists have identified the need for athletes to move between associative and dissociative states depending on the particular situation.

In my view, the ability to move between associative and dissociative states is even more important in the sport of ultrarunning. For example, we know that dissociative techniques like counting or even conversing with your fellow runner can lower perceived exertion levels (Fillingim et al. 1989; Johnson and Siegel 1992). They can be effective during the middle stages of ultramarathons when fatigue is creeping in and perception of effort needs to be minimized, thus giving rise to the sentiment of “I have a long way to go but I feel fine.” By contrast, we know that effort can be adjusted with associative strategies. For example, when you are climbing a hill early in the race and notice your breathing is becoming too labored, you can adjust your effort accordingly. Being able to move between these two strategies is a skill that should be developed in training and then strategically deployed on race day.

STRATEGIES TO ENHANCE DISSOCIATIVE FOCUS

Being able to use dissociative focus strategies can be beneficial at times when feedback, like exertion and sensations of pain, needs to be reduced to improve or sustain performance. For example, in races like the Badwater 135 and Western States 100, where an athlete's feet become a particular pain point, you can choose to intentionally redirect your focus away from your ailing feet and toward something else. However, care must be taken when utilizing these strategies, as too much or reckless dissociation can cause you to miss important perceptual cues. How many of you reading this book have tripped over a rock or hit your head on a tree branch while off in la-la land instead of focusing on the task at hand? Probably everyone (including me). My point is that dissociative strategies need to be deployed in situations where they are actually useful, and there are many where they are not.

DISSOCIATIVE STRATEGY #1: COUNT THE PURPLE THINGS. Pick your favorite color (purple in this case) and, on your next run, count the number of times you see that color on your run. It's more difficult than you think, because you have to divert attention from spotting the next rock or the next turn or remembering when you need to take a gel in order to notice fellow trail runners decked out in an array of colors. Bonus points if you do this activity with a training partner and then compare notes after the run (ultrarunners are a competitive lot).

DISSOCIATIVE STRATEGY #2: PLAN A MEAL. The next time you want to practice utilizing dissociative strategies during a run, think of an elaborate meal you want to plan for your next dinner party or special occasion. Mentally rehearse getting the ingredients out of the refrigerator, carefully measuring each of them, combining the main ingredients with the perfect amount of salt and spice, and then cooking everything to perfection. If you do this in real time, it could take hours. And that's the key: be detailed and precise, and mentally rehearse the preparation so the experience is as vivid as possible.

I like to have athletes perform either of these strategies at deliberate times during the run, normally demarked by a time or feature like a turn from one trail to the next. In this sense you force the dissociation to happen on your terms, not when your mind naturally wanders. Anybody can zone out, just ask a bored teenager during their least favorite class (English, for me). Ultimately, that's not what you are after; what you're after is the control to move to dissociation when you want to.

STRATEGIES TO ENHANCE ASSOCIATIVE FOCUS

Associate focus strategies are useful for times where you need to be in tune with what is going on with your physiology. Examples could include the beginning of a race when you are trying to pace your effort correctly, during the latter stages of a long climb as you push to the top, or while eating and drinking to ensure that you are consuming enough calories but not overloading your stomach. You can also use associative focus during fast and technical downhills to ensure precise placement of each step. In this way, many associative strategies become temporary "check-ins" with your internal physiology. Unlike dissociative strategies, they don't need to last long—just minutes for most, or maybe up to an hour if you have a very long technical downhill. Ultimately, your goal is still to be able to move between associative and dissociative strategies as the situation dictates.

ASSOCIATIVE STRATEGY #1: THE BODY SCAN. Borrowed from traditional mindfulness and meditation techniques (more on these later in the chapter), the body scan can be used as a formalized way

of doing a systems check on your entire body during a workout or race. The procedure is simple. During your next run, pick a landmark or time marker to initiate your body scan. When you get to the marker, start at your head and take note of the different sensations. Can you feel your hat moving across your forehead? What about the buckle in back? Is your ponytail swishing back and forth across your neck? Gradually and systematically move down from your head to your neck, shoulders, arms, torso, abdomen, pelvis, upper legs, lower legs, and finally your feet. As you move down, take note and appreciate each sensation you come across. As you repeat this exercise, you will become more in tune with your body's feedback. As I will explain shortly, this can be integrated into a process to help you regulate effort, become aware of minor problems before they become larger issues, and adjust your plan accordingly.

The body scan can also be used in a passive setting alongside a meditation or mindfulness practice. In fact, it's prevalent in mindfulness apps like Headspace and Calm, which are great applications for those further interested in mindfulness practice. I encourage athletes to start with a passive body scan while sitting or lying down before moving to an active body scan during a run.

ASSOCIATIVE STRATEGY #2: FOLLOW YOUR BREATH. Your breath and respiration rate tell you a tremendous amount about your effort—so much so, that during intervals your simple respiration rate can serve as a near universal proxy for heart rate and oxygen consumption. This associative strategy is specifically focused on becoming more in tune with your breath so that you can regulate exertion, particularly during harder efforts. During your next run, take a moment to direct focus to your breath. Start with where the breath is coming from. Is it coming from your nose, from your mouth, or both? As you inhale, notice your chest expanding and subsequently contracting. You can put your hand on your chest if the physical touch helps you internalize this natural rise and fall. Do not force your breath to be long and deep or shallow and fast; just let it be however long or short it needs to be depending on the level of effort. This exercise can be performed while doing intervals as well, and the focus can assist in maximizing your effort, thus simultaneously improving the quality of the workout as well as your associative technique.

Associative strategies can be used across any physical sensation, from the wholebody scan to how your muscles and joints feel. Like dissociative strategies, they should be applied deliberately, depending on your desired outcome. Learning how to purposefully shift your focus between associate and dissociative strategies gives you more opportunity to direct your mind meaningfully in important moments, both in training and racing environments.

Mindfulness

Interventions in sport tend to borrow and steal lessons from other disciplines and redirect them in pursuit of improved performance. Take, for example, a pair of compression socks or pneumatic compression device (like the Normatec boots). These devices were originally developed for patients with poor circulation stemming from diseases such as diabetes. Compression interventions had been around for a long time before someone's entrepreneurial spirit kicked in and said, "Hey, athletes need better circulation, too. Let's adapt these for a different audience." Throw in some venture capital money, some Instagram influencers, and voilà, the compression garment industry and pneumatic compression devices were born. (Side note—we discussed the effectiveness of these tools in Chapter 8.)

WHAT ABOUT MUSIC?

MANY ATHLETES CHOOSE to use music as a dissociative focus strategy and a motivational technique. Both of these would be viewed as passive ways of achieving either dissociation or motivation, but because it is a passive stimulus, using it during training is not effective for learning the skill of switching between associative and dissociative strategies. If you like to listen to music or

your favorite podcast (or even the audio version of this book), that's great. But you should still carve out time during your training to deliberately work on active interventions and learn how to switch between associative and dissociative strategies.

Mindfulness for athletes is no different. Originating in ancient Buddhism, mindfulness practices have been around for centuries as part of meditation, simultaneous connection and detachment, and religious practice. At its core, mindfulness is a belief that all perceptions stemming from your thoughts, feelings, and the inside and outside world are impermanent. Perceptions can come and go as they please, and we can alter how we react (physically and/or psychologically) to them. Bringing this core belief into the sporting realm, athletes have used mindfulness techniques to better focus attention, perceive pain, become more accepting, reduce anxiety, and improve running economy (Bishop et al. 2006; Brown and Ryan 2003; Brown, Ryan, and Creswell 2007; De Petrillo et al. 2009; Hill et al. 2020).

Despite mindfulness's religious and spiritual background and the fact that it conjures images of long-haired meditating hippies sitting around a drum circle chanting "ommmmmmmmm," the practice is not woo-woo pseudo-science fluffery at all. Neuroimaging studies performed on mindfulness practitioners have started to tease out exactly how the brain is impacted by different mindfulness interventions. Mindfulness can alter how the brain reacts to pain (Grant, Couremanche, and Rainville 2011) and affects body awareness (Hase et al. 2015; Mehling et al. 2009), both of which athletes can leverage to improve performance.

MINDFULNESS STRATEGIES

In 2004, researchers Frank Gardner and Zella Moore developed a multiphase mindfulness approach for athletes. Like any other area of athletic development, they suggest a "focus on the fundamentals" approach first, followed by a gradual ratcheting up of the mindfulness's difficulty and impact by applying it to real-world athletic scenarios. The original work, which is formally called the mindfulness-acceptance-commitment (MAC) approach to mindfulness, has been adopted and adapted by coaches, athletes, and sports psychologists to suit the needs of their particular situations. The following is my interpretation of their framework, which I have deployed with athletes. For greater detail on this work, please refer to the citation located in the references at the back of the book (Gardner and Moore 2004, 2015; Moore 2009).

STEP 1: Identifying What Matters in Competition and What Can Lead You Astray

Mindfulness for athletes does not necessarily begin with sitting on your favorite pillow in lotus position and focusing on belly breathing. Instead, take a moment and inventory what you need to be mindful about in the first place and what can distract you from those cues. For example, if you know you should be pacing your race off of your RPE, looking at your watch is a distraction. Seriously, what information salient to evaluating your RPE are you deriving from your current elevation? There can be a lot to inventory and, to be honest, it can be overwhelming. So, to start, pick a few and place the things that matter in one column and the things that can distract you from those in another.

While this is step one of your mindfulness journey, please feel free to revisit this as you move through the next three steps.

STEP 2: Becoming Acquainted with Mindfulness

Mindfulness in ultrarunning begins in a simple and unrelated setting. Acquainting oneself with mindfulness can be done during benign and routine events that are performed daily, such as cutting your food, brushing your teeth, and even folding your clothes. Additionally, a host of apps like Calm and Headspace are available to introduce athletes to basic mindfulness concepts. If you are considering mindfulness as a tool in your mental skills arsenal, I highly encourage you to start with a basic subscription to one of these apps and have the narrator lead you through some of the fundamentals. If you want to take matters into your own hands (literally, in this case), I suggest the following exercise the next time you do a load of laundry:

THINGS TO BE MINDFUL OF	THINGS THAT WILL DISTRACT YOU
Rate of perceived exertion	Looking at the pace on your watch
Internal confidence	Where you are compared to others
Taking things one mile at a time	Calculating the distance to the next aid station

TABLE 15.1 An example inventory of attentional cues to be mindful of and what thoughts can be distracting.

The next time you set out to fold your laundry, set up the room and a time so that you will be in a quiet, uninterrupted space. As you pick up each article of clothing, take time to feel the textures of fabric, noticing if your fingers move along any buttons or seams. Take a breath and notice the residual fragrance of the detergent you used. Notice where your weight is balanced on your feet. As other thoughts seep into your mind (like what time you need to pick your kids up from soccer practice or what you have left to do on that work presentation), take note of them and when they come and go. Notice when your attention has turned back to the textures and smells of your laundry task.

The point of this simple exercise is to bring awareness to all of the thoughts you have flowing in and out of your consciousness. At this point you are not taking action on them, you're merely noticing these thoughts as they come and go. You can do a similar exercise while brushing your teeth, taking a walk, sitting down on the floor in a traditional meditative setting, or any other benign task that suits you.

STEP 3: Learning to Accept

The next step in your mindfulness journey is accepting your thoughts as neither good nor bad, but as they are. Being impermanent, thoughts entering into your consciousness can either stay or you can choose to redirect them somewhere else. In any case, they are what they are. They are not good or bad (as step one in this exercise could lead you to believe). Learning to accept these thoughts without judgement is the key in this next step of mindfulness. So, if during your routine task you notice that you are being judgmental about the thoughts entering your head (i.e., "this thought is bad/counterproductive"), notice the label, let it go, accept the thought for what it is, and redirect your attention to the task at hand.

Steps two and three of this exercise can be repeated daily, and I encourage athletes to spend at least two weeks on these simple exercises before moving on to the next step.

STEP 4: Moving to the Field

Now that you have a grasp on thoughts entering your head during routine tasks, it's time to take that strategy out into the field of practice during a normal training run. For this step, you will want to start small, preferably during one of your RecoveryRuns. Pick a section of the run bounded by two landmarks approximately two to three minutes apart. As you approach the first landmark, just as you did with the laundry exercise, make note of thoughts entering your mind. Once again, your goal is not to judge or place labels at this point, just notice as the thoughts come and go.

Repeat this process on subsequent RecoveryRuns for up to two weeks.

STEP 5: Putting Things into Play

This is the step where you integrate what is beneficial and detrimental to your focus into practice. For this step, you will need to review your list from step one. Pick a day where you have a particularly demanding run or set of intervals. Either during the intervals themselves or on a climb during the run, perform your mindfulness activity as you did during the previous step. Once again, you are taking note of thoughts as they enter your head, not judging them but accepting them as they are. This time, though, for the thoughts that are beneficial to optimizing performance, direct attention toward these to take action. If you notice that your breathing rate is climbing uncontrollably during a climb, you would direct that thought to action (slowing down, in this case). If you notice you feel the need to look at your watch (which is something that would distract you from your exertion), accept that thought as neither good nor bad but for what it is, and then redirect your mind to something that is productive (feeling the sensation in your legs, for example).

This final step integrates the previous steps into a cohesive strategy where you are considering the natural thoughts entering your mind without judgement and choosing what actions to take. This entire process takes a few months to get a handle on. So, I suggest moving through these steps slowly, a few weeks at a time. The fundamentals are always important, so feel free to integrate previous steps concurrently with the step you are on.

Imagery

Imagery has become an incredibly well-established field in sports psychology and athlete preparation. In fact, there is an entire journal devoted to imagery research in sport (Journal of Imagery Research in Sport and Physical Activity). Seminal research in this area performed by Terry Orlick and John Partington in 1988 studied the mental skills of 235 Canadian athletes who participated in the 1984 Olympic Games. They found that 99 percent of these athletes performed some sort of imagery in preparation (Orlick and Partington 1988). That's a staggering percentage of adoption for any type of intervention used in sport. You would have a hard time finding such a high rate of use for a category of training aids in any other area, such as strength training, nutritional supplementation, or working with a nutritionist. And don't think for a second that Canada has a secret lock on this intervention. Imagery is used by elite athletes in every sport, worldwide. Yet despite the widespread adoption of imagery at the elite level, for amateur and age-group athletes, imagery falls way down the list of priorities.

At its root, imagery is a way of mentally rehearsing a real-life scenario (like one you would encounter in a race, for example) during training so that when an athlete actually encounters that scenario in the field of competition, the movements, reactions, sensations, and psychological responses are more familiar. Thus, even if it's the first time you've actually experienced something in real life, you can take appropriate action because you have mentally rehearsed. Imagery is a multifaceted skill that can impact many areas of your ultrarunning game. For example, as a result of mentally rehearsing what to do in a traumatic situation, you can be calmer when you get lost (trust me, if you stay in ultrarunning long enough, this will happen to you). By imagining how you hike fluidly up a 4,000-foot pass, you will be more confident when you reach the foot of that pass on race day, even if you did not experience that type of terrain in training. You can also use imagery as a way to master technical skills, like running down a scabbly descent.

DEVELOPMENT AND SEQUENCE OF IMAGERY TECHNIQUES

Like physical training, imagery training has a progression to it. Athletes initially undertaking imagery to sharpen their mental skillset should start simple and then gravitate toward more realistic, complex, and impactful imagery sessions. In this sense, you would begin imagery sessions at home, with the focus on simple scenarios encountered during training.

The setup here is simple: sit down in your favorite chair, think of the scenario that you want to work through, close your eyes, and then mentally rehearse what is going to happen. As you become more apt at taking these

homegrown imagery skills out on your training runs (minus the closing your eyes part), you can use imagery during runs to mentally rehearse race scenarios.

EXAMPLES OF IMAGERY STRATEGIES

From a practical point of view, imagery strategies for ultrarunners can fall into two different categories. In one category, ultrarunners can use imagery to mentally rehearse critical sections of a race, like smoothly going through an aid station or grinding up a difficult climb. With these scenarios, you are attempting to improve your race-day effectiveness. Going back to our earlier principle that mental skills allow you to access a greater percentage of your maximum physiological capacity, imagery facilitates that through a “been there, done that” strategy. Here’s an example:

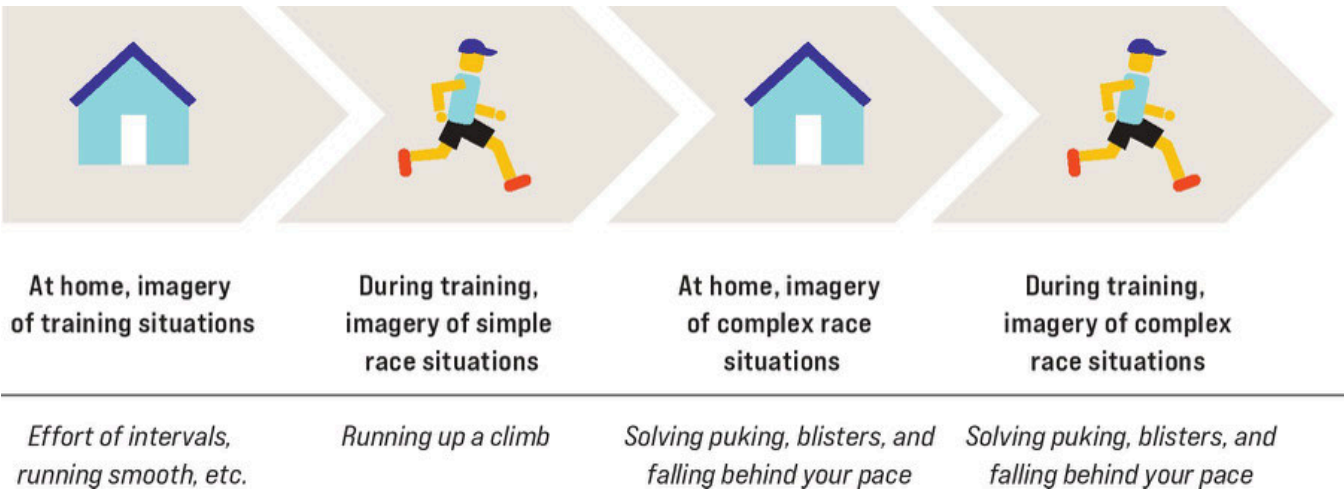


FIGURE 15.6 Sequence of imagery exercises.

Athlete Jane is racing the Hardrock 100. She trains in Minneapolis and has never been on a 14,000-foot peak before. She has access to some small local climbs with 200 feet of elevation gain, and she will be running the Silver State 50 in preparation for Hardrock. Fortunately, she has done the Silver State 50 before and knows the climbs and descents. In order to prepare for Handies Peak, which is a 14,000-foot summit in the middle of the Hardrock course, Jane wants to use imagery strategies in training so that when she encounters the climb, it is not so foreign to her.

Jane starts her imagery training at home, mentally rehearsing her next workout and how she is going to perform. Next, Jane uses her local 200-foot climb as a way of putting herself on the climbs of the Silver State 50. Each time she goes over her local climbs, she uses imagery to place herself back on the Silver State course, powerfully running and hiking each of the climbs. As Jane’s training moves along, she does some research on YouTube to find out what Handies Peak is like. She mentally rehearses that climb at home, paying specific attention to the fact that there are two distinct pitches, separated by a talus field. The terrain moves from sharp rocks to loose gravel as the climb gets higher, and she imagines how her feet will feel as the surface changes. Finally, Jane moves these imagery sessions into the field of practice during training runs on her local 200-foot hill as well as during the Silver State 50.

The example above can be extrapolated to fit almost any race situation. The point of the whole exercise is to gradually move from simple imagery sessions that are relatively contrived to complex ones that will represent the real world.

The second category of imagery strategies is a form of contingency planning for when things inevitably go awry. As I have mentioned multiple times during the course of this book, if you stay in ultrarunning for any appreciable amount of time, it's not a matter of if, but rather when you will experience a catastrophic series of events that have the potential to take you out of a race you have trained so hard for. Particularly for things like 200-mile events and longer stage races, the duration and difficulty of the events result in unforeseen challenges that can throw even the most accomplished ultrarunners off their game. So, for athletes in this situation, I recommend imagery sessions that feature the absolute worst possible combination of things that can happen in a race. The exercise looks something like this:

STEP 1: Write Down the Worst Possible Race Situation You Can Imagine

We've all heard the horror stories of athletes breaking a headlamp, running out of an aid station without some critical piece of gear, getting lost, or bonking so hard that they swear they were abducted by aliens. Your first step in this exercise is to pick three or four of these horror stories, write them down, and then pick a point in your upcoming race where this cosmic misalignment of fortunes might actually happen all at once. Dialogue as much of the fictional narrative as you can, in as much detail as possible. Your situation might look like the following:

I am running down Hope Pass, forty-four miles into the Leadville Trail 100. Suddenly I catch my toe on a rock and tumble down the trail. A few runners stop to help, and I notice my knee is split open and my hands have deep cuts from sliding down the scree. The bottles in the pack I was wearing broke in the whole ordeal, as did my trekking poles. As I continue to walk down the descent, I notice that all of my calories were somehow jettisoned out of my pack. I continue on toward the Winfield aid station without any fluid or food for the next two hours, and suddenly my stomach starts to feel nauseous. I throw up before getting to the aid station. As I enter the aid station, I realize that my crew has not made it there yet. So now I am only fifty miles into a 100-mile race, I have no crew, I have no way of carrying fluids, I am nauseated and bleeding from my knees, and I have a 2,000-foot climb right in front of me that I have to conquer.

STEP 2: Read Your Story. What Are You Going to Do about It?

Next, mentally rehearse what you are going to do to get out of that situation and keep running down the trail. Be specific; go step by step. For the example above, put yourself in the Winfield aid station. Realize that there are other crews and volunteers there to help. Go through each of your ailments and what you are going to do about it, step by step, one by one, as specifically as possible. If in your scenario you get patched up by medical personnel at the scene, visualize the color of their hair, make up a name for each of them, and imagine the conversation you will have. If in your scenario you have to borrow a pack from a pacer, what model was it and what were the colorways? The point of the exercise is to be as vivid and real as possible, rather than simply creating a counterchecklist to your series of mishaps described earlier. Your imagery exercise should end with you successfully continuing down the trail—perhaps with some bandages and a pack you borrowed from a volunteer, and without your poles, but you are continuing down the trail nonetheless, having turned what could have been a DNF into an epic comeback.

There's a lot going on in that scenario, but I have seen it happen exactly like that, as well as countless similar iterations. When faced with such scenarios, some athletes find the wherewithal to continue down the trail, and others throw in the towel. Using imagery to dig yourself out of a hole in practice can be the turning point for rescuing your race in real life.

Self-Talk

“Keep pushing.” “You’ve got this!” “You’ve worked so hard for this.” “I am going to make it to the finish.”

All of these statements are forms of self-talk used by ultrarunners. Self-talk can take the form of instructional (“Plant your poles in line with your feet”) or motivational (“You’ve got this”). In each of these cases, the talk should be matched to the desired outcome. If you need a pick-me-up, “You’ve got this” is going to be worth a whole lot more than “Plant your poles in line with your feet.” To facilitate this alignment, as well as to give some framework for utilizing self-talk, psychology researcher Antonis Hatzigeorgiadis and his colleagues developed the IMPACT approach to self-talk.

- **IDENTIFY** what you want to achieve: Self-talk, like all training interventions, needs to have a specific goal in mind.
- **MATCH** self-talk to needs: What self-talk cues are going to facilitate the goals you identified?
- **PRACTICE** cues with consistency: In training, go try out your cues.
- **ASCERTAIN** what worked: Evaluate what cues worked and what cues did not work. Refine your plan accordingly.
- **CREATE A PLAN**: Create your self-talk plan from what worked in training. If you are using self-talk for a race, draw out what self-talk strategies might be helpful during different stages of the race.
- **TRAIN**: Like anything in life, you have to practice to get good!

If you have not caught on by now, the structure of designing self-talk interventions mirrors that of mindfulness, imagery, and training at large. In essence you are finding a self-talk plan that suits your needs, stress-testing that plan, and then training with that plan for eventual use during a race. Of particular note with self-talk, the type of self-talk you develop has to be specific to the outcome you want to achieve. Similar to the instructional vs motivational self-talk example, in an ultramarathon you are going to want different cues at different points of a race because your desired outcomes will change. For example, “You’ve got this” might be a poor self-talk cue in the beginning of a 100K race because it encourages you to dig deep or go harder when your goal at that point should be to conserve energy and relax. I’ve drawn out some of these scenarios in Table 15.2.

The remaining steps of the IMPACT self-talk strategy are relatively self-explanatory. You are testing out your self-talk strategies in training, seeing what works, modifying along the way, and ultimately determining a framework for what you will use during a race.

IDENTIFY WHAT YOU WANT TO ACHIEVE	MATCH SELF-TALK
Keep pushing when the race gets hard	“You’ve trained hard enough,” “You’ve got this”
Run your own race	“Relax, focus on your effort”
Maximize effort during a training session	“Keep pushing, almost there”
Pole-strike effectively	“Plant your pole firmly, follow all the way through”

TABLE 15.2 Matching self-talk strategies to what you want to achieve.

Purpose Through WHY

The societal, healthcare, and business responses to the COVID-19 pandemic will be studied for decades to come. The widespread disruption of established patterns and routines was totally unanticipated, and in the endurance community it tested the role of events for keeping athletes engaged. With most events cancelled, athletes lost the familiar guideposts they used to navigate through their training and competition seasons. Many athletes completely lost their way and went from being disciplined and goal-oriented to scattered and totally unmotivated in a matter of weeks. For some, this led to such frustration that they ended up throwing in the towel altogether. Maybe they stayed generally active, but they dropped any pretense of “training.” Other athletes never skipped a beat, even as their race calendars went up in smoke, gyms and offices closed, and stay-at-home orders were enacted. These athletes continued to train, had no problem with their daily 4:30 a.m. wake-up calls, continued to plug in their headlamps, and carried on with the daily grind. The difference between the two is that for the latter group, running served a purpose greater than collecting finisher medals and belt buckles.

Certainly, racing can be a powerful source of motivation. There’s nothing wrong with competition as a part of the reason—or WHY—behind your commitment to running. But as we saw during the pandemic, there is a danger to overemphasizing races when considering the purpose running serves for you. Even in non-pandemic conditions, a strong purpose adds a turbocharger to your motivation engine fueled by competition.

Fortunately, I can almost guarantee you already have some underlying WHY for running besides the finish lines and belt buckles. Through more than twenty years of coaching runners, I can only recall a handful who only ran to race (and they didn’t stick around in ultrarunning very long). You have to have a purpose, because running is not very glorious most of the time. There are times when it downright sucks. So, how do you find your WHY? I adapted the following framework from the renowned Simon Sinek, author of *Start with Why*, and include a personal example of going through the exercise myself.

STEP 1: Write Down All the Things You Do Consistently

Even gurus have to go grocery shopping. At the other end of the spectrum, people who feel like anonymous cogs in the machine often do things of great meaning—even if they don’t realize it. You and I are somewhere in the vast middle between these extremes. Our lives contain a long list of basic tasks, but my guess is that you do some meaningful things on a consistent basis. Your first step in developing your individual WHY is to take an inventory of the things you voluntarily do on a regular basis. “Getting out of bed” or “fixing dinner for the kids” are both important but won’t quite cut it at this stage. Think of the things that you look forward to, don’t put off for later, and derive satisfaction from. Bonus points for activities you don’t post on Instagram, that no one will know about, and for which no one will give you a lick of credit. These are activities you do for you. Don’t try to assign meaning to them yet, just get them down on paper.

Figure 15.7 offers a simplified record from my brainstorming session for this step. I know that without fail I will run, coach, mentor other coaches, and build and help maintain trails in my local area. At the time I did the exercise, I actually had no idea why those were the things I did consistently and without much prompting. Nonetheless, I still do them. My guess is that many of you reading this book will have “running” as one of your consistent and nonnegotiable activities. If it is, then great. If not, don’t worry, there’s nothing wrong with you. This is just the first step in the process.

THINGS I DO CONSISTENTLY



FIGURE 15.7 Step one in finding your why.

STEP 2: Find the Threads That Are Intertwined with Your Actions

The next step is to figure out the themes your most consistent activities have in common. This will likely take a while, so be patient. It took me a month to get through this step. I recommend setting aside time each week to come back to the exercise, review the connections you have already established, and search for the ones you haven't made quite yet. In my example, it became clear to me that after I worked on a trail, mentored a coach, or went for a run, something or someone was likely better off because of my effort. The themes of "betterment" and "improvement" started to resonate with me (as highlighted below).

When you do this for yourself, avoid simple truisms like "they make me feel good," "it's motivating," or "I love it." Dig deeper than superficial feelings and emotions. If it makes you feel good, why? If you love it, what about it do you love?

THINGS I DO CONSISTENTLY

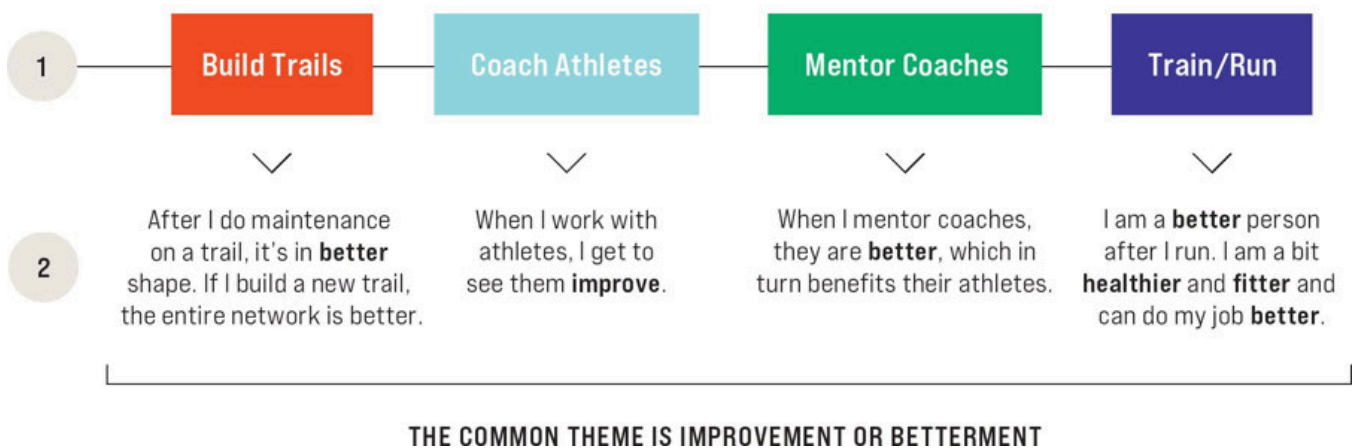


FIGURE 15.8 Step two in finding your why.

STEP 3: Phone a Friend

Sometimes we are not great at creating accurate depictions of ourselves. You know yourself well, and hopefully a bit better after the first two steps of this exercise, but at this stage some outside perspective is going to help. Up to this point you have been using self-evaluation, which can be biased toward telling the story of who we want to be or feel we should be rather than who we actually are at the present moment. This is normal, particularly for athletes, who are a naturally aspirational group.

So, now's the time to bring in one of your running friends, preferably one who knows you and is not afraid to tell it like it is. Don't waste time searching for the perfect person, just pick someone you can bounce ideas off of and who will offer up an external perspective. Their first role is to vet your "Things I do consistently" list. (Come on now, do you really volunteer at the soup kitchen all that often?) Next, encourage your friend to push you and poke holes in the themes you've identified as connections between your consistent activities. Outside perspective is helpful for separating what sounds good from what's real.

STEP 4: Craft Your Personal WHY

Now that you have your list of things you do without fail and the themes that connect them, it's time to craft your personal WHY. Simple is better, so try to keep it to one sentence and stick to the point. While you will have many interconnected themes between your consistent actions, choose the one (or two at the most) that are the most powerful. This is not a time for multilevel, choose-your-own-adventure-style plans. You need something concrete, actionable, and simple.

For me, that WHY boiled down to: "To leave things better than I found them." Yes, it's something you have heard hundreds of times, and admittedly it's a bit cliché. But that's fine, because it resonates with me. I can go for a run, work on a local trail, or help out a coach or athlete and know that at the end of the day my actions were aligned with my personal WHY.

STEP 5: Drilling Your WHY Back to the Beginning

Remember the PBM way back in the beginning of this chapter? The anchor points of this model are rooted in motivation and perception of effort. Your WHY is inherently tied to and reinforces your motivation. This matters in your day-to-day activities but is put under the magnifying glass during difficult athletic endeavors such as hard training and ultramarathon racing. The stronger your WHY, the more purposeful your motivation is, which in turn drives the entirety of the training process.

Sequence of Mental Skills Development

All endurance training is chronic. Measurable and meaningful improvement happens over the course of weeks and months rather than days or hours. Mental skills are similar. Just like you wouldn't train for a 100-mile event by doing a single ten-mile run, it is unrealistic to think that a single round of imagery, one five-minute mindfulness session, or a few sentences of self-talk will have a meaningful effect on your mental skills. It takes time and deliberate, repetitive practice to hone mental skills. There is also a preferred sequence to folding these into your training routine (much as I alluded to in the imagery strategies sections) that will enhance their effectiveness.

Earlier in this book, I presented a periodization scheme that begins with training that is least specific to the event and then moves deliberately to training that is most specific to the event. For most athletes training for ultramarathons, this leads to doing low-volume/high-intensity work early and then high-volume/low-intensity work as the race draws near. Designing training in this fashion optimally prepares you for whatever the race will throw at you. Physical training also gets harder over the course of time (remember the principle of progression?). While you first might start out with a five by three-minute set of RunningIntervals, during the next training phase that targets this intensity range you might be able to handle a greater load (seven by three, for example) and perhaps at a higher absolute intensity. Mental skills are no different. They can flow from general to specific and have a similar "loading" effect, where each subsequent task becomes gradually more challenging. You can even

overlay your mental skills progression on your LRP in the “notes” or “other” section, if you like, which is something I do with my athletes to demonstrate how something as benign as sitting at home thinking about how you are going to pace your next interval set eventually leads to conquering something as arduous as Hope Pass on the Leadville course.

Below is a schematic that represents how the various mental skills can flow from the beginning to the end of a season, culminating in a race, as well an example of how each mental skill can “load” (increasing the difficulty and thus ultimate effectiveness). Starting with general associative and dissociative strategies, you lay the foundation of how to focus inward and outward and how to intentionally move between these states. During the planning process you may not know the details of the associative and dissociative thoughts that will be effective, you just know you will be using this time to hone the ability to shift focus on and off of yourself. Then, as you introduce mindfulness and self-talk, you will add meaning to what you are focusing on, creating greater depth and context. Finally, as you move to imagery, you will integrate all of these steps to mentally rehearse scenarios so you are ready to take actions and make decisions that impact your performance.

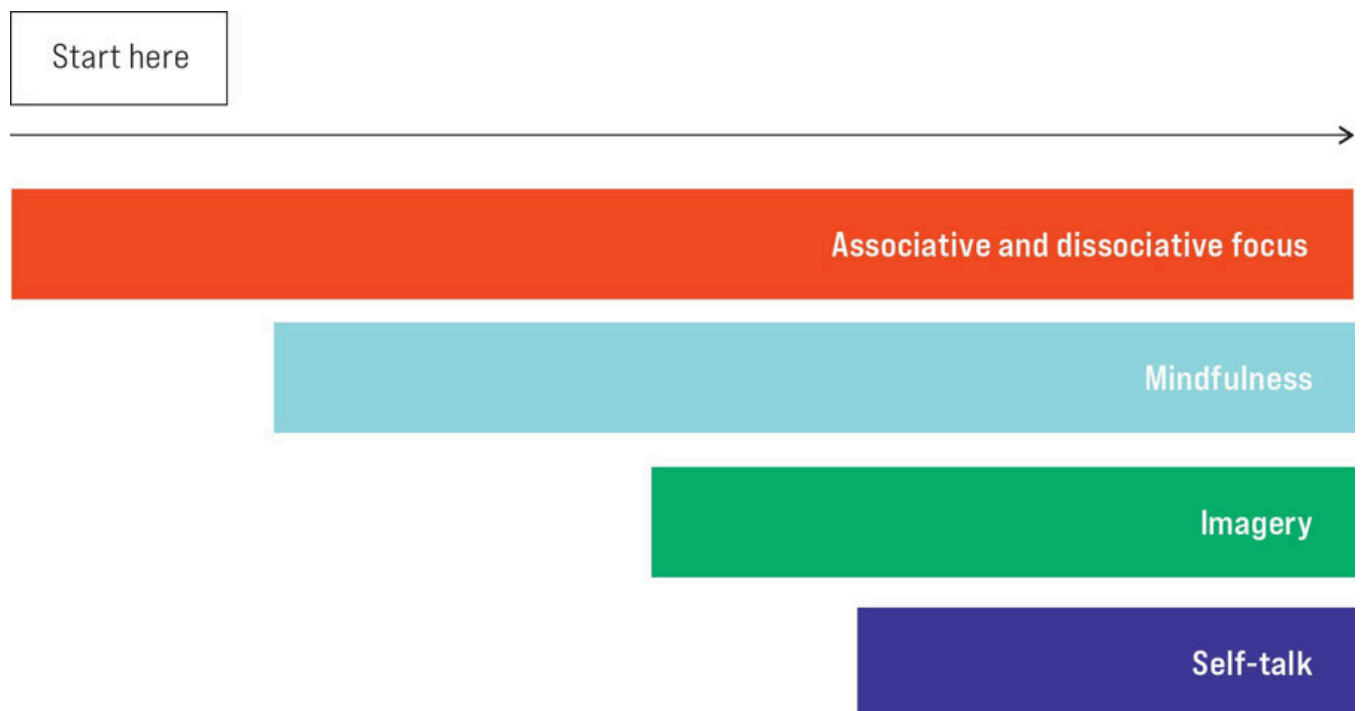


FIGURE 15.9 Where to start and how to incorporate mental skills.

KEY POINTS FOR MENTAL SKILLS TRAINING

Ultrarunning is a very physical sport, but your mental skills are still important.

Your physiological capacity determines the upper limit of performance, while your mental skills dictate how much of your physical capacity you ultimately use.

Mental skills interventions have solid mechanistic rationale that has been demonstrated in research. However, translation to performance has been equivocal.

The psychobiological model of fatigue states that how you interact with your body's physiology will regulate performance.

Associative and dissociative strategies can both be used by ultrarunners to impact performance. Ultrarunners should learn to use and move between both of these types of strategies.

A mindfulness practice can be useful to better manage the many impermanent thoughts flowing in and out of your head, homing in on those that are beneficial for performance while allowing those that are not beneficial to pass.

Imagery can be used to mentally rehearse training and race scenarios in advance so that when the time comes, action is automated.

Imagery techniques can be particularly effective in preventing a DNF when confronted with many confounding factors.

Self-talk is another mental skills technique that can be used for motivation or for instructional purposes. Self-talk should be aligned specifically with the goals it is intended to achieve.

Finding your larger purpose, or your WHY, can ground your everyday and athletic actions beyond outcome goals such as finishing a race or earning your next belt buckle.

You should deploy mental skills interventions in a logical sequence for optimal performance. That sequence is: (1) associative and dissociative focus, (2) mindfulness, (3) imagery, and (4) self-talk.



